

Computer Networks and Network Security

1. Definition of a Computer Network

A computer network is a system consisting of two or more electronic devices that are connected together to exchange information, share files, and access common resources such as printers and internet connections. These devices may include computers, smartphones, tablets, printers, or any other digital equipment capable of data communication. For example, when a file is sent from a laptop to another computer using Wi-Fi, this represents a simple computer network in action.



Key Characteristics or Benefits of Networking

1. Connectivity: Allows devices to communicate and exchange data efficiently.
2. Resource Sharing: Enables multiple users to share hardware resources and services.
3. Scalability: Networks can grow easily by adding new devices.
4. Reliability: Ensures consistent and stable data transfer.

2. Types of Computer Networks

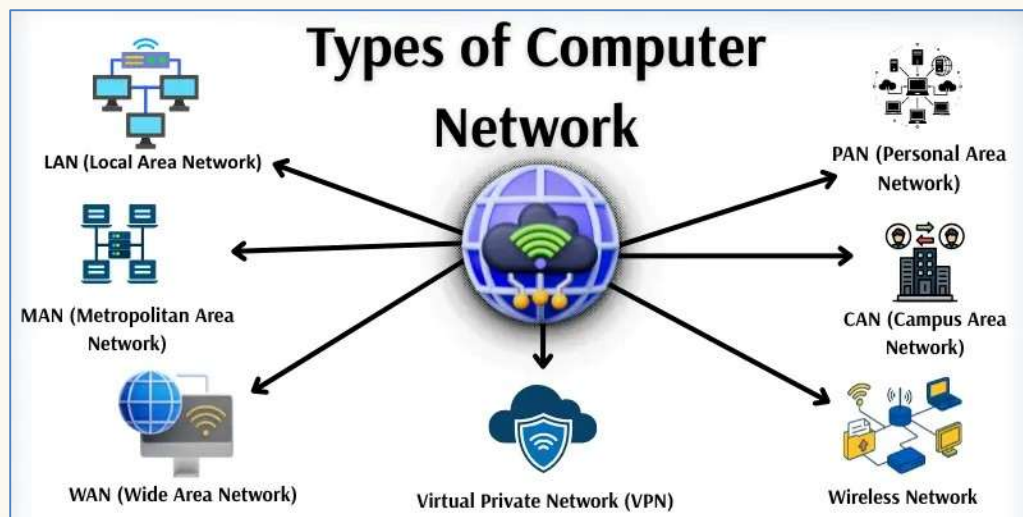
Computer networks can be classified based on their geographical coverage, ownership, and purpose. The four major types are:

PAN – Personal Area Network

LAN – Local Area Network

MAN – Metropolitan Area Network

WAN – Wide Area Network



3. Personal Area Network (PAN)

A PAN is the smallest type of computer network and is primarily intended for individual use. It connects devices like laptops, smartphones, and printers within a short distance.

Examples:

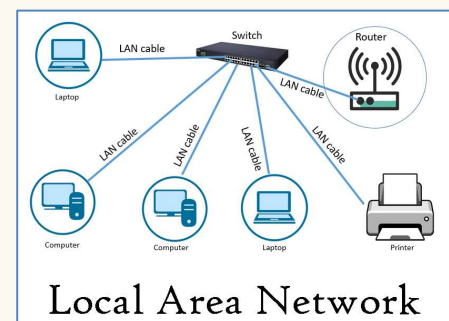
- Bluetooth, USB.



4. Local Area Network (LAN)

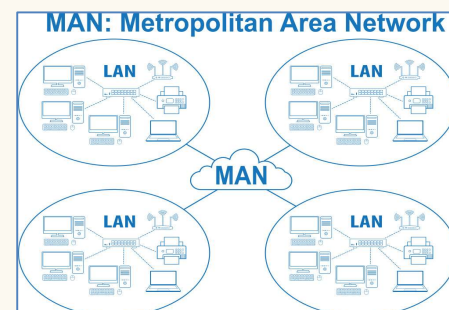
A LAN connects multiple devices within a limited area such as a school, office, or organization. It allows users to share files, applications, printers, and internet connections efficiently.

Examples: Wi-Fi networks in homes or universities.



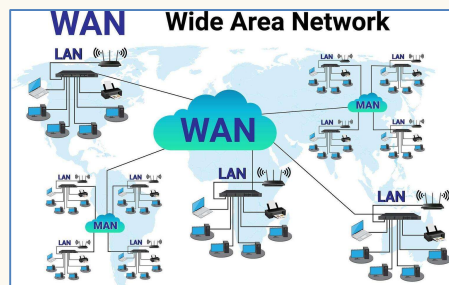
5. Metropolitan Area Network (MAN)

A MAN spans across a city or a large geographical area, connecting several LANs together. It's ideal for universities, government institutions, or city-wide communication systems.



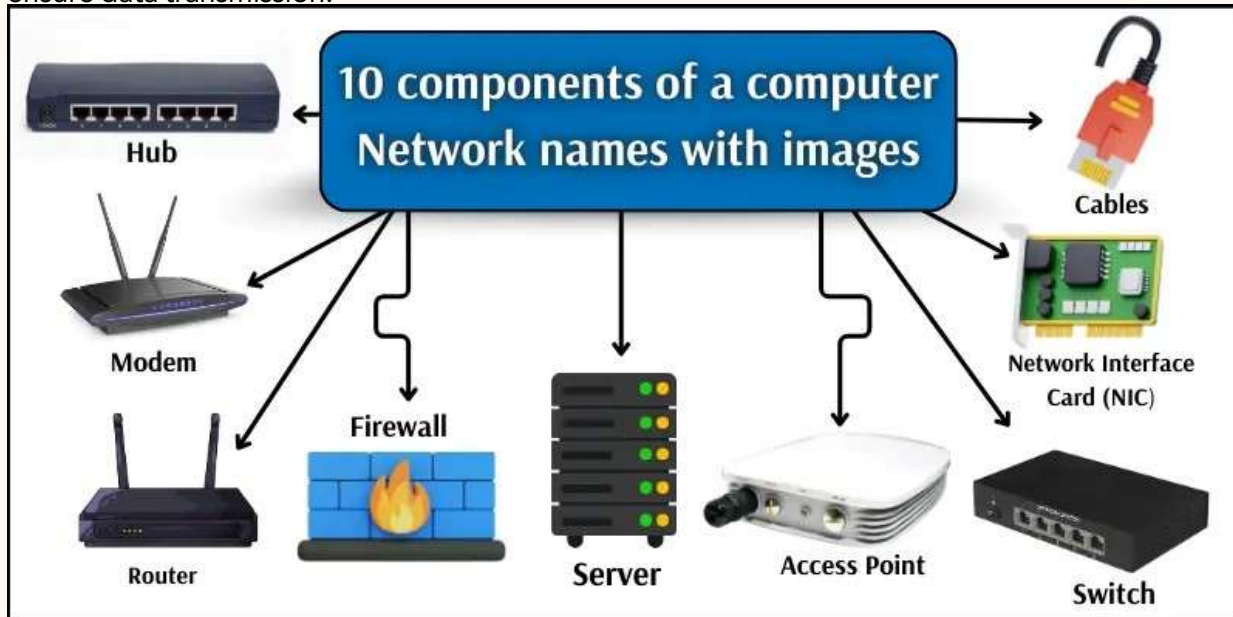
6. Wide Area Network (WAN)

A WAN is the largest type of network, connecting devices and LANs across countries or continents. The Internet is the best-known example of a WAN.



7. Basic Components of a Computer Network

A computer network is built from both hardware and software components that work together to ensure data transmission.



Hardware Components:

- Computer (Node)
- Network Interface Card (NIC)
- Switch
- Router
- Modem
- Transmission Media

Software Components:

- Network Operating System (NOS)
- Protocols (TCP/IP)
- Network Management Tools

8. Network Security: Definition and Importance

Network Security refers to the collection of technologies, procedures, and best practices used to protect computer networks from unauthorized access, misuse, malfunction, or destruction.

9. Common Network Threats

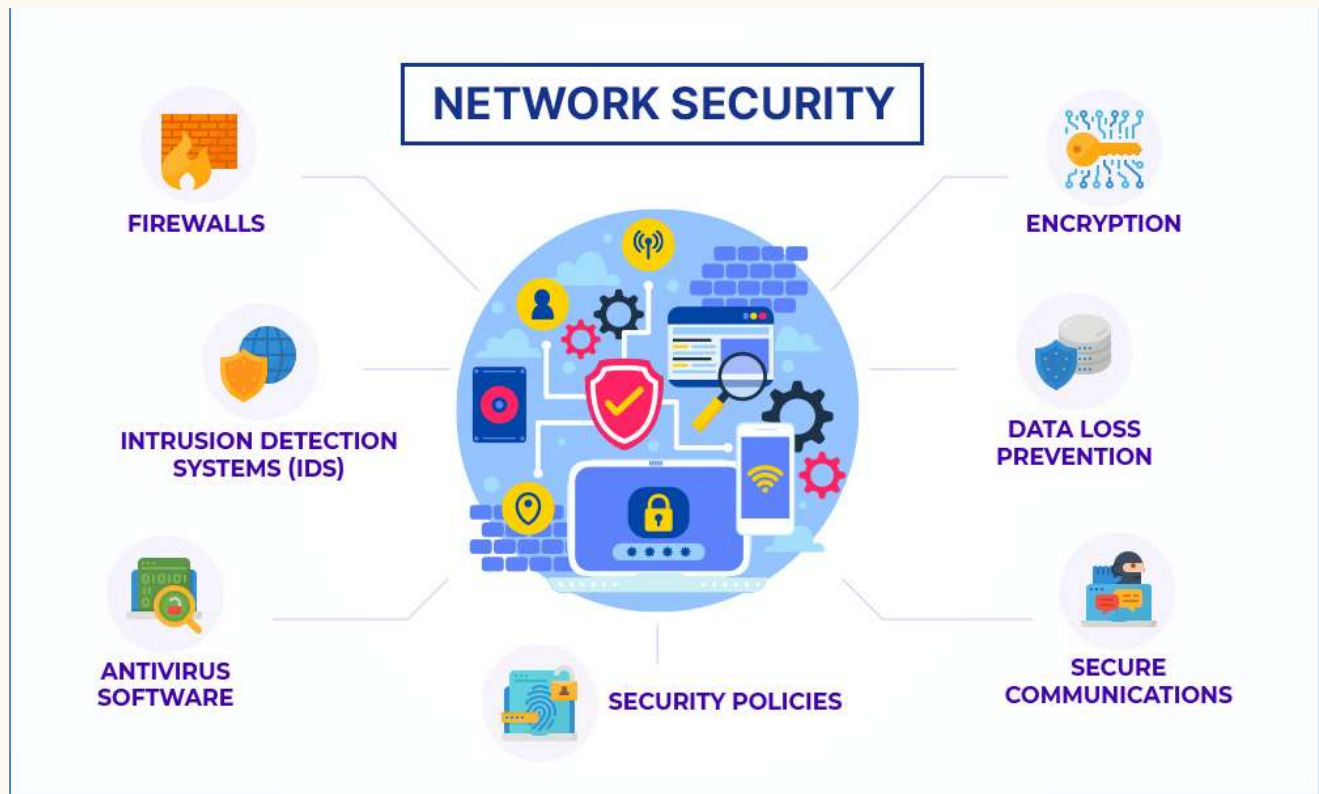
Network threats include any activity that can harm, disrupt, or gain unauthorized access to systems.

- A. Malware: Software designed to damage or control computer systems.
- B. Phishing: Tricking users into revealing confidential data.
- C. Denial of Service (DoS): Overloading servers with traffic.
- D. Unauthorized Access: Gaining entry without permission.
- E. Social Engineering: Manipulating people to reveal private information.

10. Basic Network Security Measures

Security Measures:

- Strong Passwords
- Two-Factor Authentication (2FA)
- Firewalls
- Antivirus & Anti-malware
- Encryption
- Regular Software Updates



Cybersecurity Best Practices for Users

- Do not share personal or login information.
- Avoid clicking unknown links or suspicious attachments.
- Check for 'https://' in website addresses.
- Log out from public computers.
- Regularly back up important data.

11. Summary

- Computer networks enable communication and data exchange.
- Networks vary in size: PAN, LAN, MAN, WAN.
- Components include routers, switches, NICs, and firewalls.
- Security ensures confidentiality, integrity, and availability.
- Awareness and protection are key to cybersecurity.